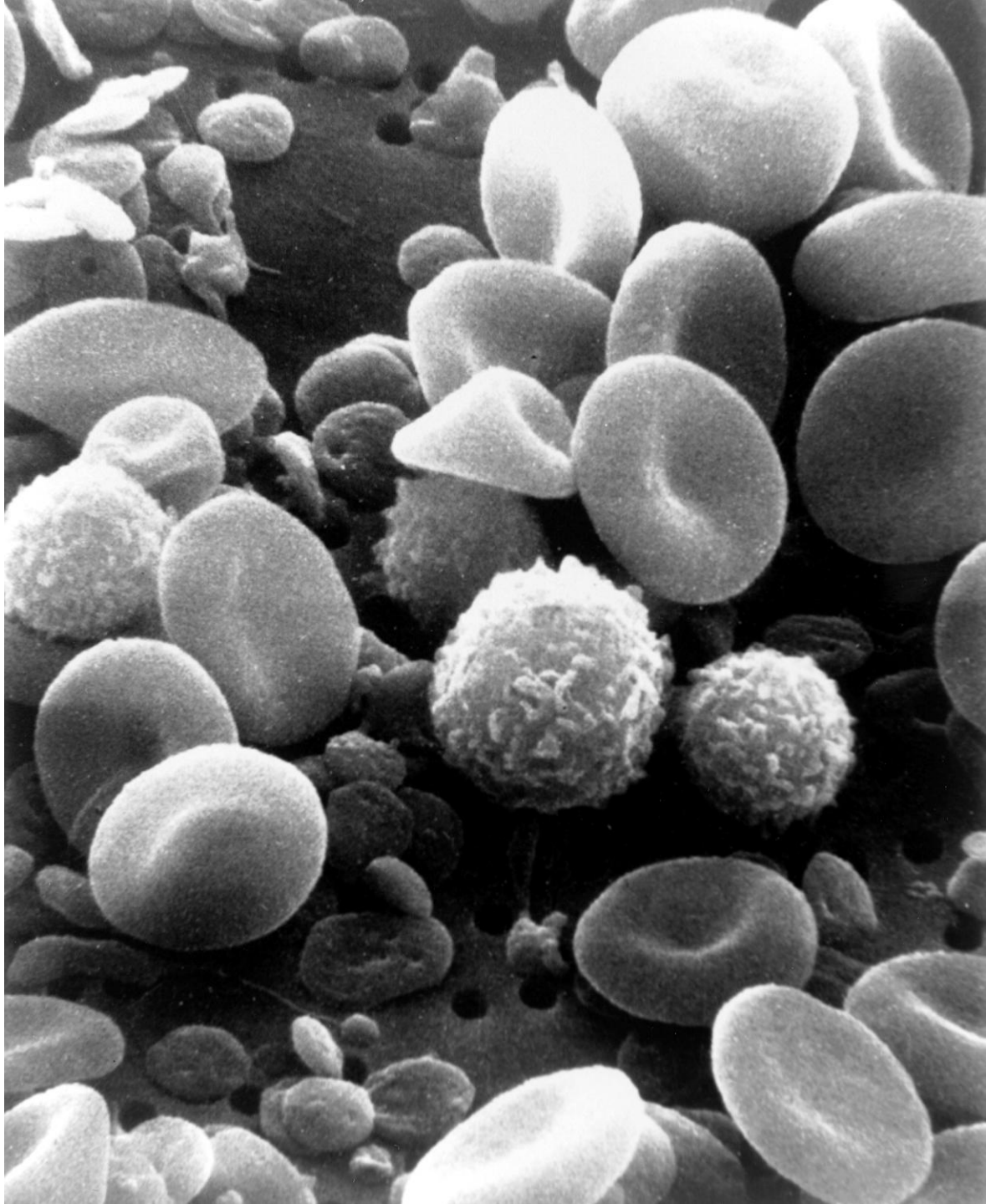
DP Biology – C3.2

Defence against disease

- How does the body detect pathogens and fight infection?
- What affects how common a disease is in a population?

SL and HL. Textbook pp. 517–547.

Image: human blood cells, scanning electron micrograph. National Cancer Institute, public domain.

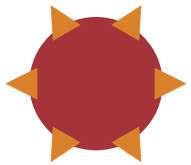


Objectives for each class

Class	Statements	By the end of the class, you can...
1	C3.2.1–5	Name the groups of pathogen and explain how skin, mucous membranes, clotting and phagocytes defend the body
2	C3.2.6–12	Explain how lymphocytes make antibodies and memory cells, and how HIV leads to AIDS
3	C3.2.13–18	Explain antibiotics and resistance, zoonoses, vaccines and herd immunity, and evaluate pandemic data

Three lines of defence

C3.2.1–5

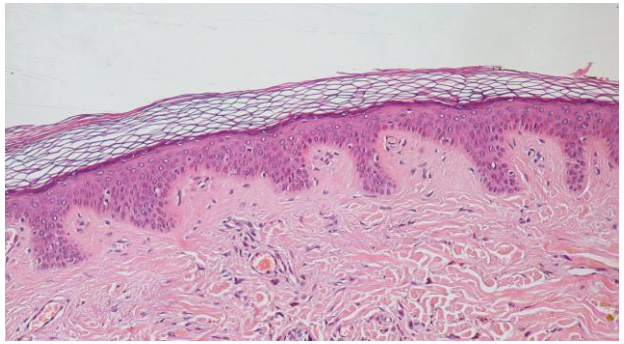


pathogen

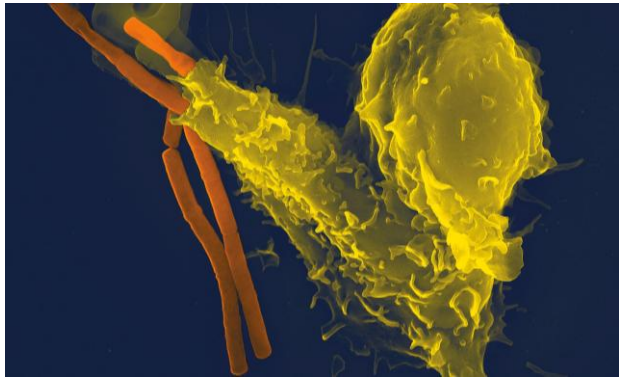
1 Barriers
skin, mucous membranes, clotting

2 Innate system
phagocytes: fast, not specific

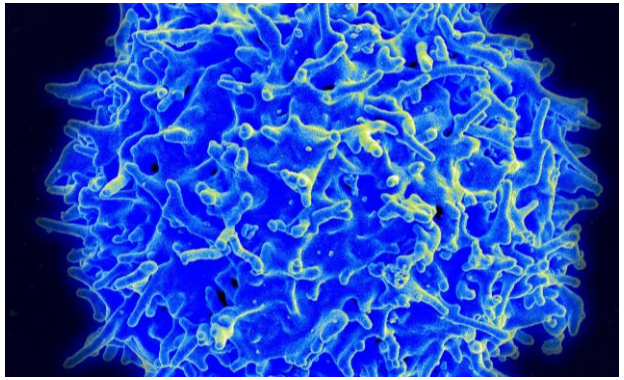
3 Adaptive system
lymphocytes: slow first time, specific, with memory



Skin section. Kilbad, public domain.



Neutrophil engulfing bacteria. Volker Brinkmann, CC BY 2.5.



T-lymphocyte. NIAID/NIH, public domain.

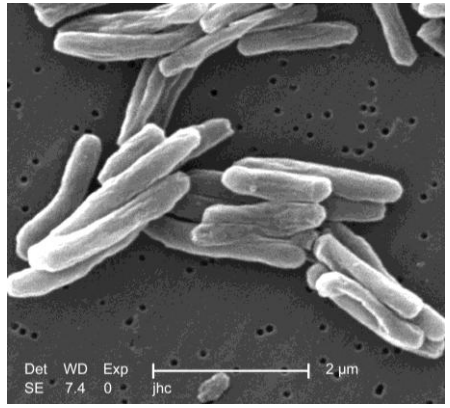
Key idea

Each line catches what gets past the one before. Only the third line is **specific** to one pathogen, and only it **improves with experience**, because it keeps memory cells.

Pathogens: five groups

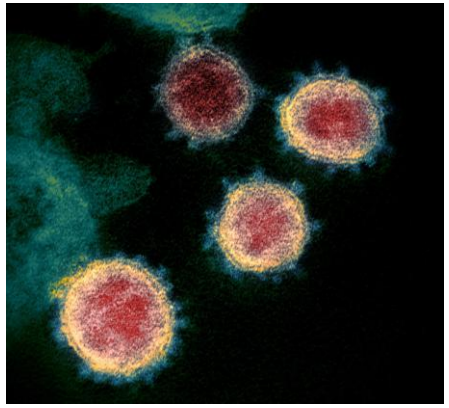
C3.2.1

A **pathogen** is an organism or virus that causes disease in its **host**. Diseases it causes are **infectious**: they can pass from host to host.



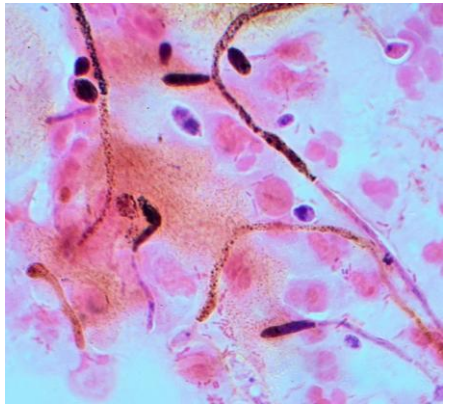
Bacteria

Prokaryotes. Cholera, tuberculosis



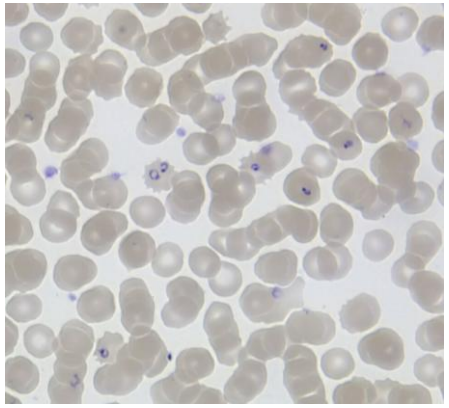
Viruses

Not cells; reproduce only in host cells. Flu, HIV



Fungi

Eukaryotes with chitin walls. Thrush



Protists

Single-celled eukaryotes. Malaria



Parasitic worms

Multicellular. Schistosomiasis

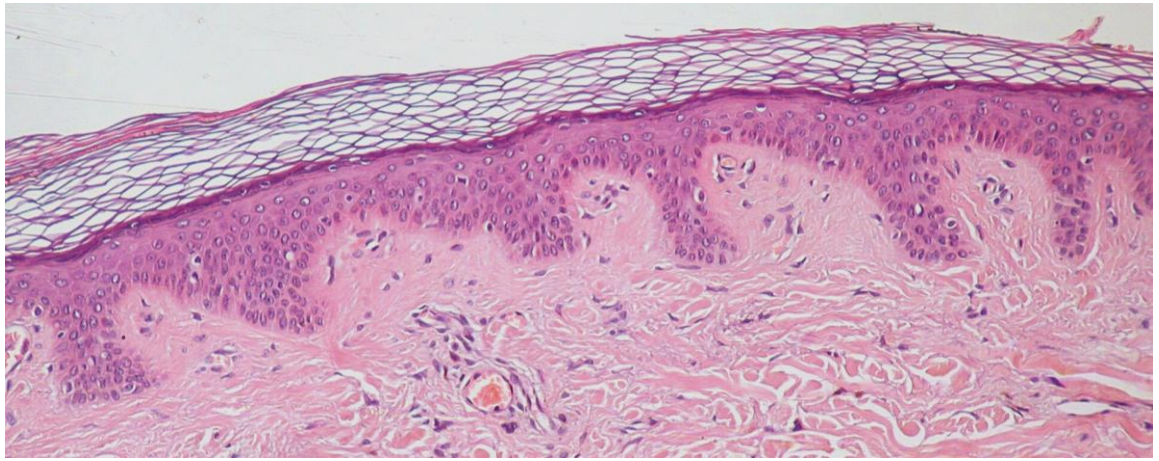
Images: M. tuberculosis (CDC, PD); SARS-CoV-2 (NIAID, CC BY 2.0); Candida (Graham Beards, CC BY-SA 4.0); Plasmodium in blood (M. Zahniser, PD); Schistosoma (D. Williams, PD).

Not infectious

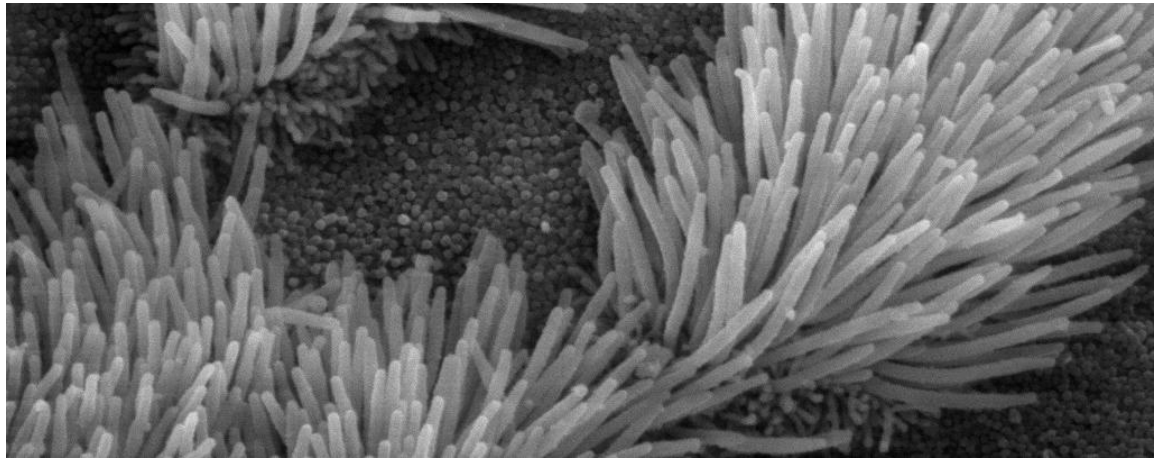
Cancer, malnutrition and inherited conditions are diseases, but they do not pass between people. No archaea are known to cause human disease.

Skin and mucous membranes

C3.2.2



Skin: the pale top layer is dead, keratin-filled cells. Kilbad, public domain.



Cilia lining an airway (SEM). Charles Daghlian, public domain.

Physical: tough outer layer of dead cells packed with **keratin**, sealed by lipid.

Chemical: sebum and sweat keep the surface acidic, and **lysozyme** damages bacterial walls.

Line the airways, gut and other openings: thin and living, so easier to cross.

Sticky **mucus** traps microbes; **cilia** sweep it up to the throat to be swallowed. Stomach acid kills many.

Think

Why do smokers, whose cilia are damaged, get more chest infections?

Blood clotting seals a cut

C3.2.3

Cut: platelets and damaged tissue release **clotting factors**

↓

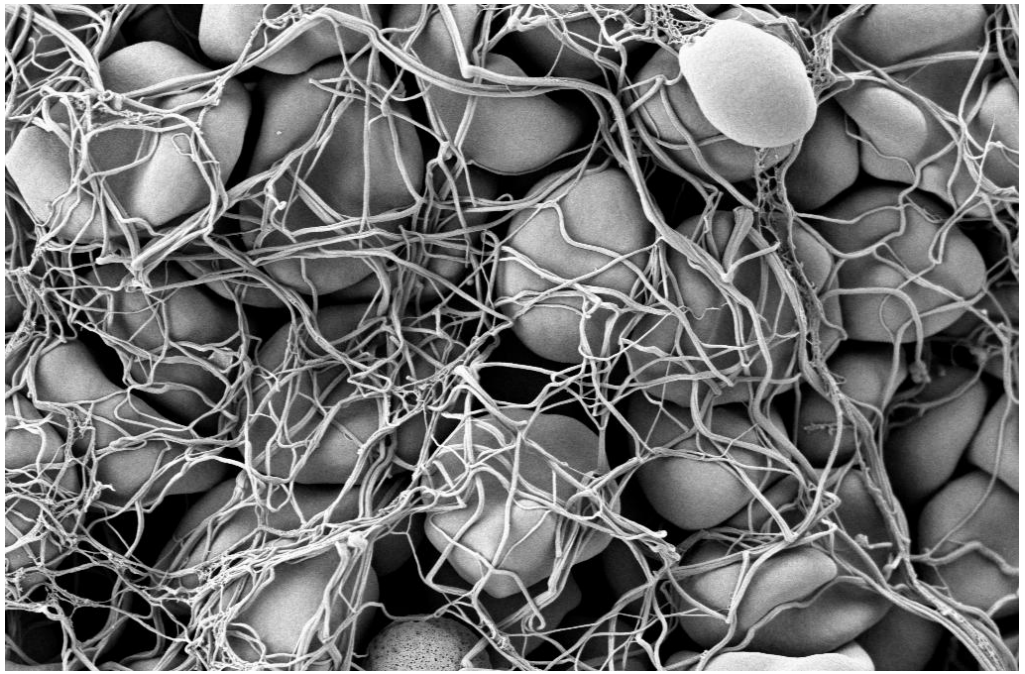
Prothrombin → **thrombin** (an active enzyme)

↓

Thrombin: soluble **fibrinogen** → insoluble **fibrin**

↓

Fibrin fibres form a mesh that traps red cells: a clot



Red blood cells trapped in fibrin (SEM). Fuzis, CC BY-SA 4.0.

Why it matters

The clot stops blood loss and **keeps pathogens out** until the skin heals. It is a cascade: each step activates the next, so a few factors quickly make a lot of fibrin, but only where there is damage.

Innate and adaptive systems

C3.2.4

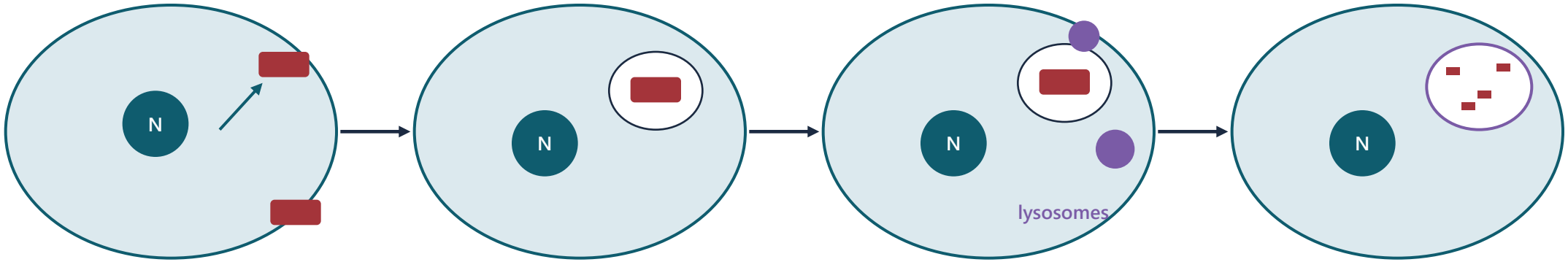
	Innate system	Adaptive system
Recognises	Broad groups of pathogen	One specific pathogen
Speed	Hours	Days the first time
Second time	Same as the first	Faster and stronger
Over a lifetime	Does not change	Builds memory
Cells to know	Phagocytes	Lymphocytes: B-cells, helper T-cells

In short

You are born with the innate system. The adaptive system is shaped by the pathogens you meet: that is why you usually get chickenpox only once.

Phagocytes engulf and digest

C3.2.5

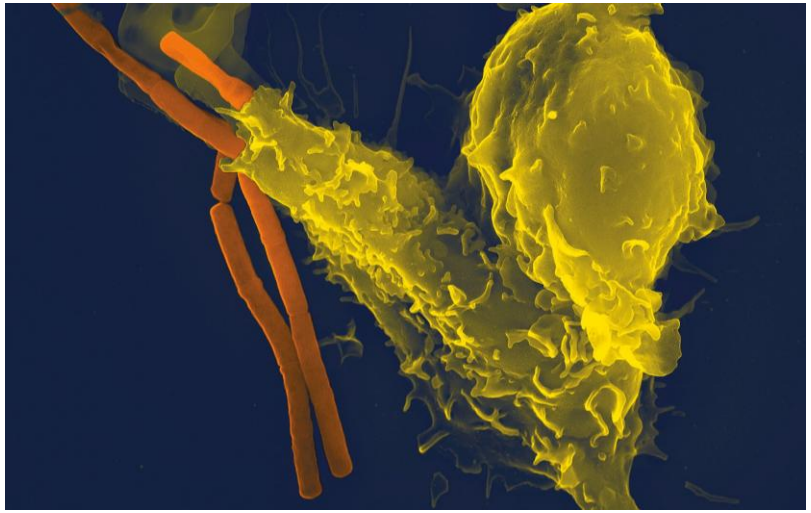


1 Moves to the infection by amoeboid movement

2 Engulfs the bacterium by endocytosis

3 A lysosome fuses with the vesicle

4 Enzymes digest the bacterium



Neutrophil (yellow) engulfing bacteria (orange), coloured SEM. Volker Brinkmann, CC BY 2.5.

Phagocytes squeeze out of capillaries and move to the infected tissue. They recognise pathogens as foreign, take them in, and digest them with **lysosome** enzymes.

They attack **any** pathogen of a broad type, so they are part of the innate system.

Working with data: clotting time

C3.2.3

Time for blood plasma samples to clot after a clotting factor was added. These data are invented for practice; the pattern is realistic.

Sample	Clotting time / s
Normal plasma	12
Plasma with fibrinogen removed	no clot after 300
Plasma with extra thrombin added	5
Plasma from a patient with haemophilia	45
Normal plasma at 10 °C	40

1. Explain why plasma without fibrinogen does not clot.
2. Explain why extra thrombin shortens the clotting time.
3. Haemophilia means one clotting factor is missing. Suggest why clotting is slow, not absent.
4. Suggest why clotting is slower at 10 °C.

Check-in: class 1

C3.2.1–5

1. State what makes a disease infectious.
2. Explain one physical and one chemical way that skin keeps pathogens out.
3. Describe how mucus and cilia protect the airways.
4. Outline how a clot is formed, naming thrombin and fibrin.
5. Give two differences between innate and adaptive immunity.
6. Describe how a phagocyte destroys a bacterium.

How to use this

Answer without your notes, then check against slides 4–8.

Lymphocytes: B-cells and helper T-cells

C3.2.6

B-cells

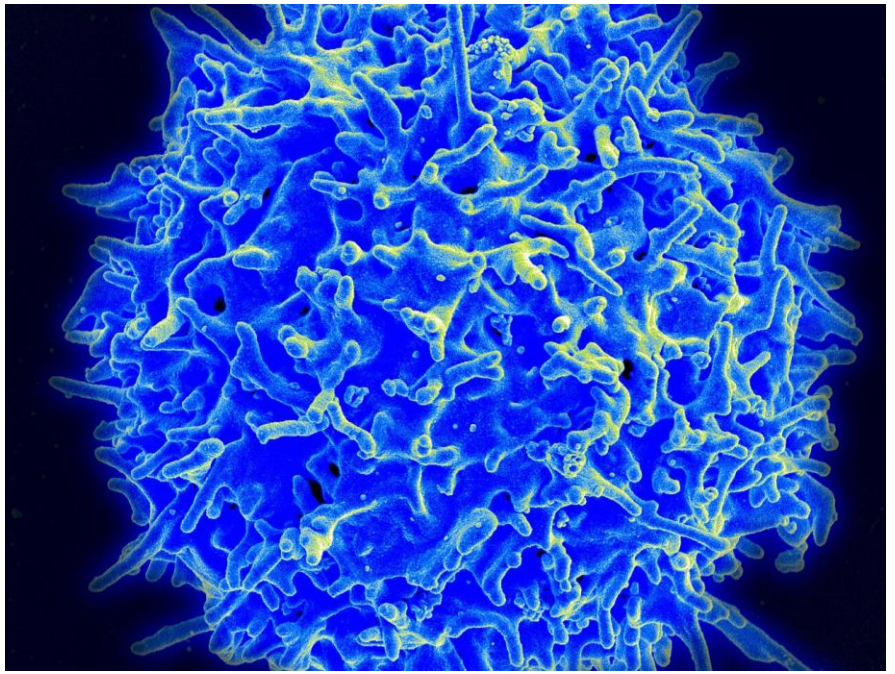
Each type makes **one** specific antibody.

Once activated they become plasma cells, which secrete it.

Helper T-cells

Each type recognises **one** antigen.

They make no antibody: they **activate** the B-cells that match.

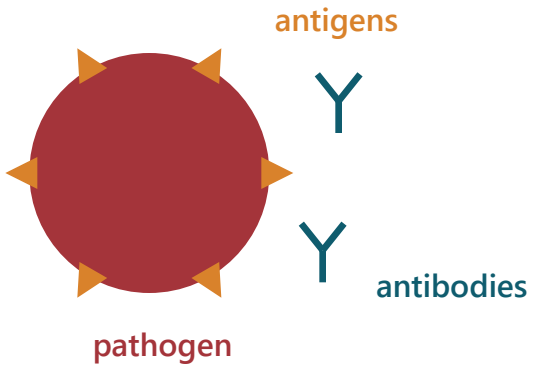


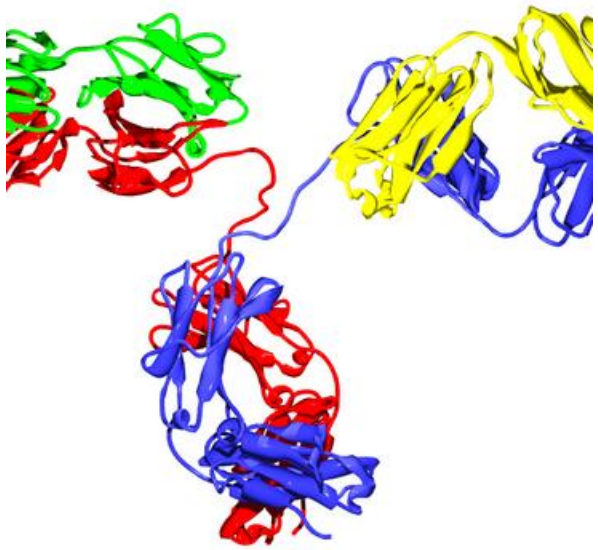
Human T-lymphocyte (SEM). NIAID/NIH, public domain.

Many types, few of each

There are millions of types of lymphocyte, so almost any antigen is matched by some. But only a few cells of each type exist, far too few to make enough antibody: the matching ones must be selected and multiplied first.

Antigens and antibodies





An antibody, about 10 nm tall (PDB 1IGT). TimVickers, public domain.

An **antigen** is a molecule recognised as foreign, usually a protein or glycoprotein on the pathogen surface.

An **antibody** is a Y-shaped protein. The tips of its arms fit one antigen exactly.

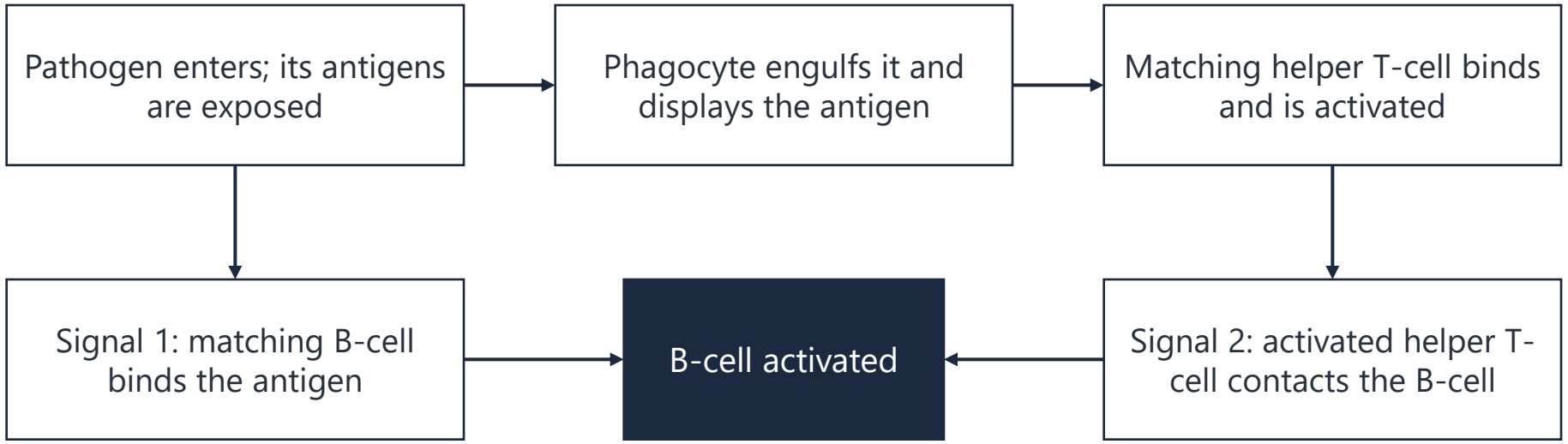
Blood group	Antigens on red cells	Antibodies in plasma
A	A	Anti-B
B	B	Anti-A
O	None	Anti-A and anti-B

Blood groups

Red cell antigens trigger antibodies too. Giving the wrong group makes the donor cells clump.

Activating a B-cell: two signals

C3.2.8



Rule

Both signals must come from the **same antigen**.

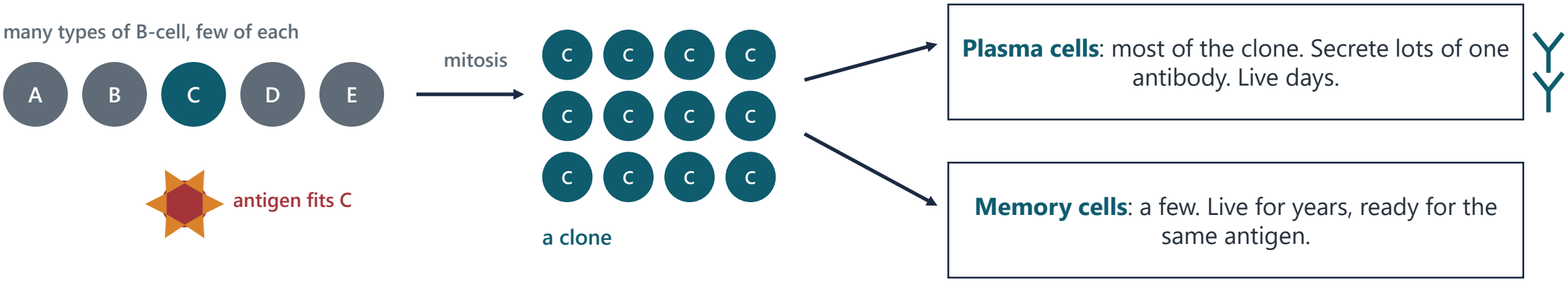
Two signals act as a safety check: a B-cell only starts making antibody when a helper T-cell confirms the antigen is foreign. Helper T-cells make no antibody themselves.

Then

The activated B-cell divides (next slide). This is why the first response takes days.

Clones of plasma cells

C3.2.9



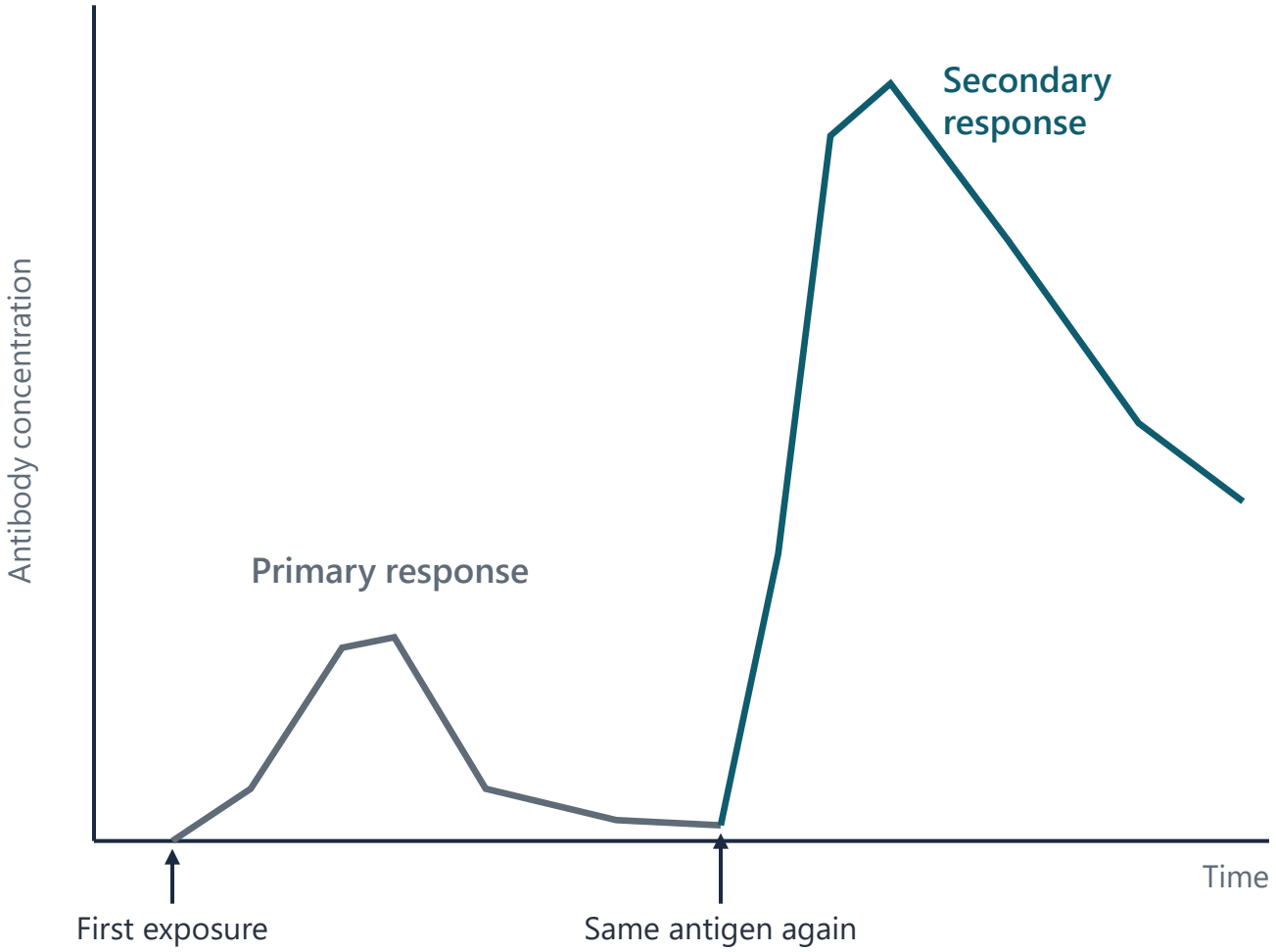
Only the B-cell whose antibody fits the antigen is **selected**. It divides by mitosis into a **clone** of identical cells.

Plasma cells are packed with rough ER, the site of antibody synthesis. Thousands of them all secrete the same antibody.

Name

This is **clonal selection**: select the matching cell, then clone it. It solves the "few of each" problem from slide 11.

Memory cells give immunity



Schematic, not to scale.

Immunity is being able to get rid of an infection before it makes you ill.

Plasma cells die off, but **memory cells** stay. On a second exposure they divide at once, so antibody is made **sooner**, **faster** and to a **higher level**.

Think

Which steps from slides 13 and 14 are skipped or shortened the second time?

How HIV spreads

Passes in body fluids

- Unprotected sex
- Shared needles
- Mother to baby, at birth or in breast milk
- Unscreened blood or transplants

Does not pass by

- Sharing cups, plates or towels
- Hugging or shaking hands
- Saliva, tears or sweat
- Insect bites



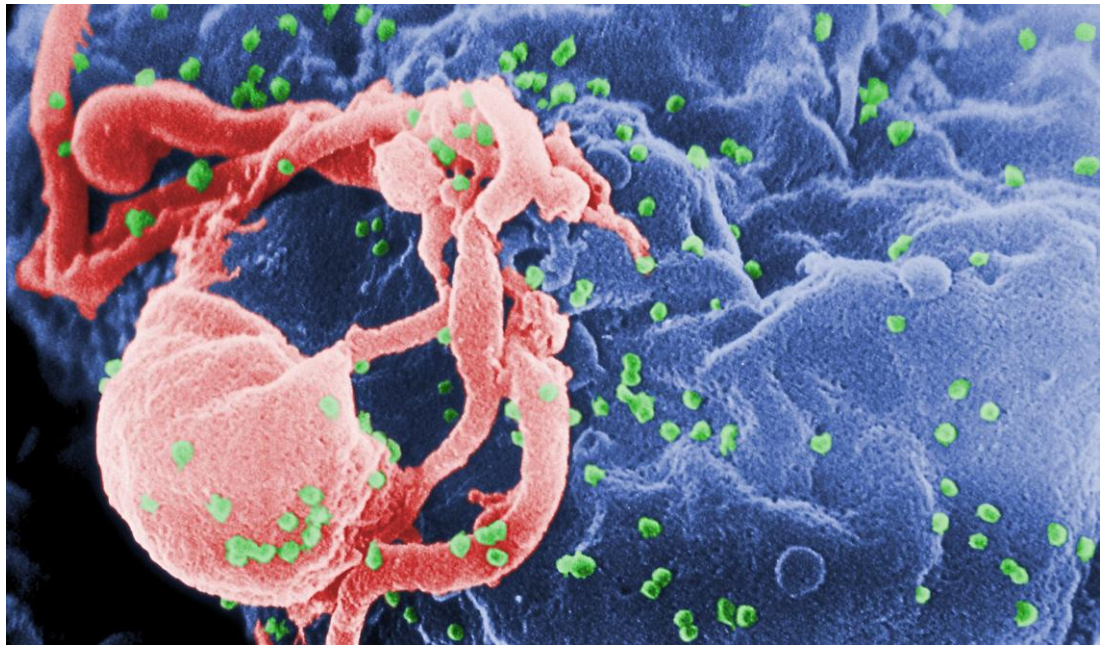
Blood donation. Armed Services Blood Program, public domain.

So

Condoms, clean needles, screening donated blood and treating pregnant mothers all block transmission. Knowing how HIV does **not** spread also reduces stigma. Write "HIV is transmitted": AIDS is the condition, not the virus.

HIV infects helper T-cells

1. HIV binds to **CD4** receptors on a helper T-cell
2. Its RNA enters the cell
3. **Reverse transcriptase** copies the RNA into DNA
4. The DNA joins a host chromosome
5. New viruses are made; infected cells die



HIV (green) budding from a lymphocyte, coloured SEM. CDC, public domain.

fewer helper T-cells → fewer B-cells activated → too little antibody → other infections take hold: AIDS

AIDS (acquired immunodeficiency syndrome) is diagnosed by these **opportunistic** infections, such as tuberculosis and pneumonia, often years after infection.

Working with data: HIV over ten years

C3.2.12

Helper T-cell count and virus level in one untreated patient. These data are invented for practice; the pattern matches real infections.

Years after infection	Helper T-cells / per mm^3	Virus level
0	1000	none
0.25	550	very high
1	780	low
4	520	low
7	300	rising
10	90	very high

1. Describe the change in helper T-cells in the first year.
2. Suggest why the virus level falls after a few months.
3. A count below 200 per mm^3 marks AIDS. Estimate when this patient reached it.
4. Explain why the patient then catches infections easily.

Check-in: class 2

C3.2.6–12

1. Distinguish between an antigen and an antibody.
2. Describe the two signals needed to activate a B-cell.
3. Explain why an activated B-cell must divide before enough antibody is made.
4. Compare plasma cells and memory cells.
5. List three ways HIV can be transmitted.
6. Explain how loss of helper T-cells leads to AIDS.

How to use this

Draw the two-signal diagram from memory first, then answer.

Antibiotics: bacteria only

C3.2.13

Process blocked	In bacteria	In human cells
Cell wall synthesis	Peptidoglycan wall weakens; cell bursts	No cell wall
Protein synthesis	70S ribosomes blocked	80S ribosomes unaffected
Enzymes for DNA or metabolism	Bacterial enzymes blocked	Different enzymes



Penicillium, source of penicillin. Crulina 98, CC BY-SA 3.0.

Why safe

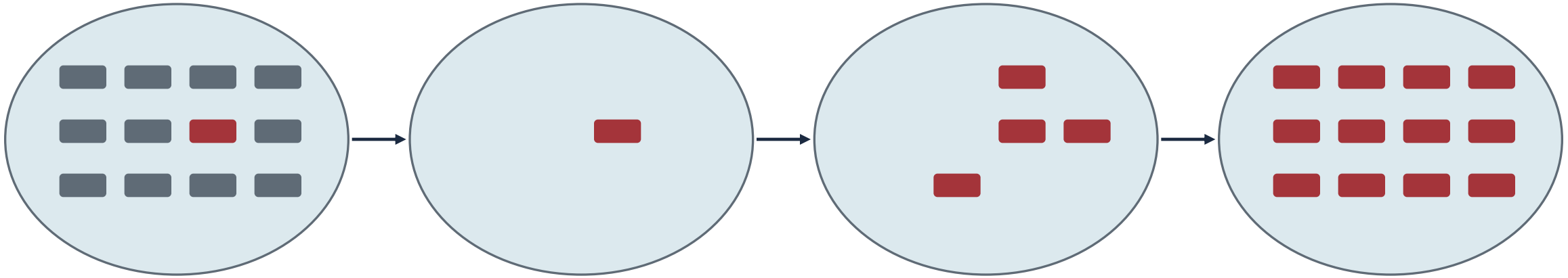
Antibiotics hit structures that **prokaryotes** have and our cells do not, so they kill bacteria inside a patient without harming the patient.

Why useless on viruses

Viruses have **no wall, no ribosomes and no metabolism** of their own: they use the host cell. Antivirals target viral steps, such as reverse transcriptase.

Resistance evolves by natural selection

C3.2.14



1 Variation: one bacterium has a resistance mutation

2 Antibiotic kills the others

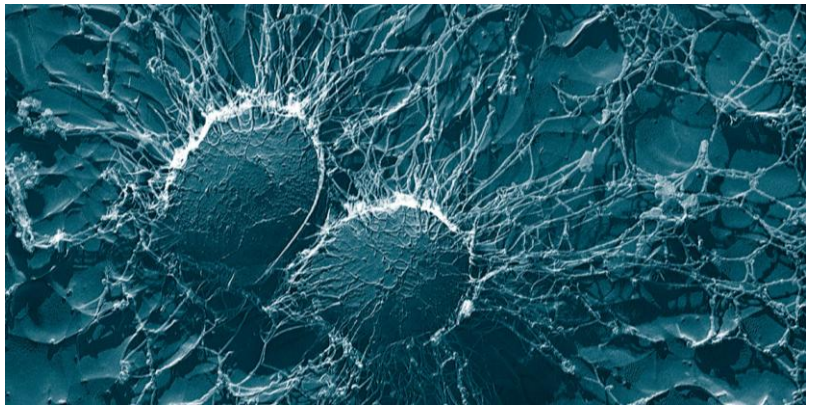
3 Resistant cell multiplies

4 Resistant population

red = resistant; grey = killed by the antibiotic

The antibiotic does not cause the mutation: it **selects** cells that already have it.

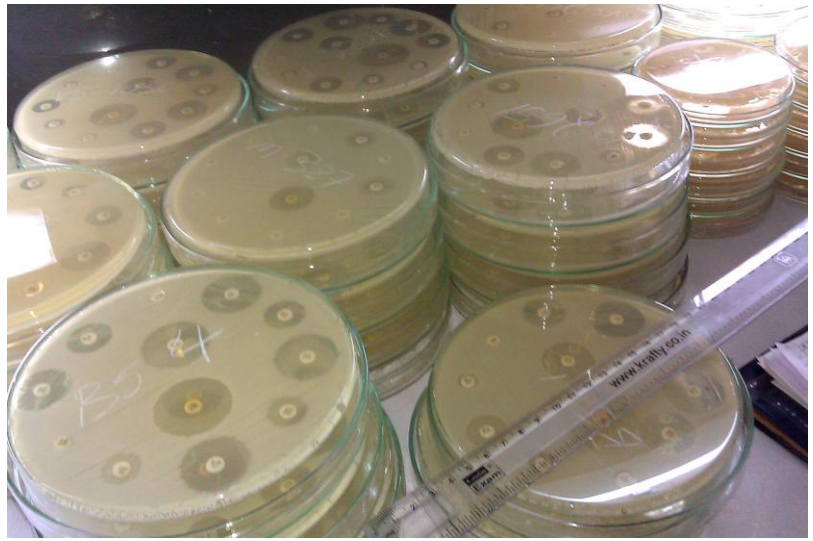
Resistance genes are often on **plasmids**, which can pass between bacteria, so one strain can collect resistance to several antibiotics.



Staphylococcus aureus, which includes MRSA (SEM). USDA, public domain.

Testing and slowing resistance

C3.2.14



Disc diffusion test. Microrao, CC BY-SA 4.0.

Each disc holds an antibiotic. A **clear zone** means the bacteria were killed; growth up to the disc means **resistance**.

Who	How to slow resistance
Doctors	Prescribe only for bacterial infections
Patients	Finish the course; never share or save
Hospitals	Strict hygiene to stop resistant strains spreading
Farms	No routine use to speed animal growth

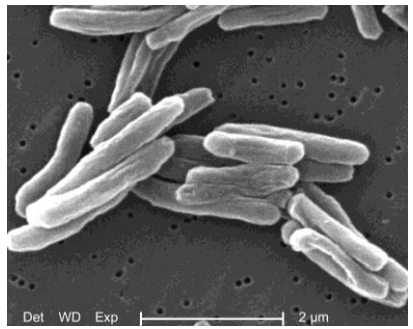
Why it matters

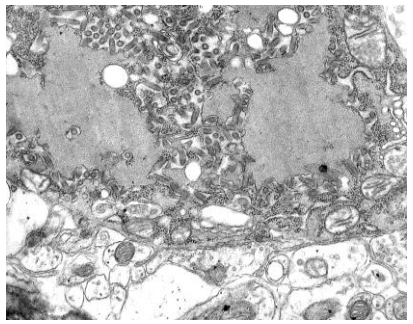
Every use selects for resistance. Few new antibiotics are being found, so the ones we have must be used carefully.

Zoonoses: from animals to humans

C3.2.15

Disease	Pathogen	Animal source	Route to humans
Tuberculosis	Bacterium (M. bovis)	Cattle	Raw milk, infected meat
Rabies	Virus	Dogs, bats	Bite; saliva in a wound
Japanese encephalitis	Virus	Pigs, wading birds	Mosquito bite
COVID-19	Virus (SARS-CoV-2)	Probably bats	Now spreads person to person







M. tuberculosis (CDC, PD); rabies virus (CDC/F. Murphy, PD); Culex mosquito (S. Hinkley and K. Walker, Museum Victoria, CC BY 3.0 AU).

Why more zoonoses?

More contact with wild and farmed animals, habitat loss and fast travel give pathogens more chances to cross to humans and spread.

Vaccines: immunity without the illness

Vaccine: antigen, or mRNA or DNA coding for one

↓

Primary response, but no disease

↓

Memory cells remain

↓

Real pathogen meets a fast secondary response



COVID-19 vaccination. US federal government, public domain.

What a vaccine contains

A weakened or inactivated pathogen, purified antigens, or nucleic acid that makes our cells produce the antigen. All give memory cells against **one** pathogen. Smallpox was wiped out by vaccination by 1980.

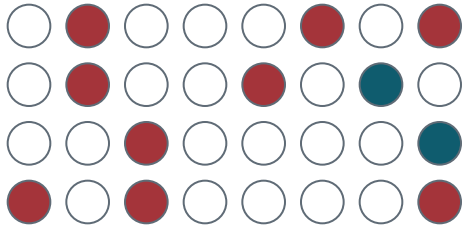
Herd immunity

C3.2.17

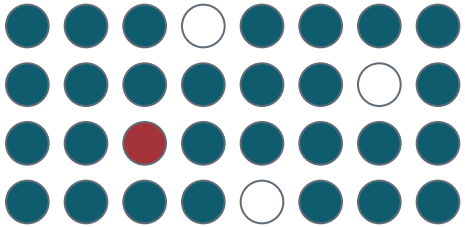


Brooklyn Bridge walkway. EazyDee107, CC0.

Few immune: infection spreads



Most immune: infection stalls



○ Not immune ● Infected ● Immune

Very contagious diseases need more people immune: about 95% for measles.

Key idea

When most people are immune, an infected person rarely meets anyone who can catch it, so the chain of transmission breaks. This also protects people who cannot be vaccinated, such as newborns and people with weak immune systems.

Two calculations for pandemic data

C3.2.18

Percentage change

(new − original) / original × 100

One value over time, or a vaccine against a placebo.
Trial: 162 cases with placebo, 8 with vaccine (groups of similar size):
(8 − 162) / 162 × 100 = -95%, so 95% fewer cases.

Percentage difference

difference / average × 100

Two values at the same time.
Invented example: 150 and 90 cases per 100 000 in two countries:
(150 − 90) / 120 × 100 = 50%

Common mistake

Check which one is asked for. Percentage change: divide by the **original** value. Percentage difference: divide by the **average** of the two. Compare cases **per 100 000 people**, not raw totals, when countries differ in size.

Working with data: evaluating pandemic figures

C3.2.18

New COVID-19 cases per 100 000 people per week in two countries. These data are invented for practice; the pattern is realistic.

Week	Country X	Country Y
1	40	35
2	95	50
3	210	70
4	180	66
5	120	60

1. Calculate the percentage change in X from week 1 to week 3.
2. Calculate the percentage difference between X and Y in week 3.
3. Suggest three reasons, other than real infection rates, why the countries differ.

Real data

Repeat with two real countries on the WHO COVID-19 dashboard (data.who.int/dashboards/covid19), using one month of data.

Common mistakes to avoid

Not this	But this
Antibiotics kill viruses.	They act on bacterial walls, ribosomes and enzymes, which viruses lack.
The antibiotic causes the resistance mutation.	The mutation was already there; the antibiotic selects it.
Helper T-cells make antibodies.	Plasma cells do; helper T-cells activate B-cells.
AIDS is transmitted.	HIV is transmitted; AIDS is the condition it leads to.
Antigens are made by the body to fight pathogens.	Antigens are on the pathogen; antibodies are made by the body.
Fibrinogen forms the clot.	Thrombin turns soluble fibrinogen into insoluble fibrin.

The whole topic on one slide

C3.2.1–18

Part	Key ideas
Pathogens	Bacteria, viruses, fungi, protists, worms cause infectious disease
Barriers	Skin (keratin, sebum, lysozyme), mucus and cilia, clotting (thrombin, fibrin)
Innate	Phagocytes: endocytosis and lysosome enzymes; not specific
Adaptive	Antigen → two signals → clone → plasma cells (antibody) and memory cells
HIV	Body fluids; destroys helper T-cells; AIDS
Medicine	Antibiotics hit bacteria only; resistance by natural selection
Populations	Zoonoses; vaccines give memory; herd immunity; per-capita data

Check that you can...

1. **Outline** the groups of pathogen and what makes a disease infectious
2. **Explain** how skin, mucous membranes and clotting keep pathogens out
3. **Distinguish** between innate and adaptive immunity
4. **Describe** how phagocytes destroy pathogens
5. **Explain** how B-cells are activated, cloned and give lasting immunity
6. **Explain** how HIV spreads and leads to AIDS
7. **Explain** why antibiotics fail on viruses and how resistance evolves
8. **Explain** zoonoses, vaccines and herd immunity, and **calculate** percentage change and difference

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Distinguish between an antigen and an antibody.
2. A patient asks for antibiotics for flu. In one sentence, explain why they will not help.
3. Explain why a second infection by the same pathogen is usually milder than the first.

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